

Properties of Convolution

Chapter 7

June 4, 2026

Central idea

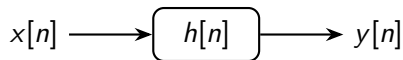
For a linear time-invariant (LTI) system, the impulse response completely determines how every input signal is transformed. Convolution is the mathematics that connects the input, the impulse response, and the output.

Convolution as the System Law

Discrete-time LTI relationship

$$y[n] = x[n] * h[n] = \sum_{k=-\infty}^{\infty} x[k] h[n - k].$$

- $x[n]$: input signal.
- $h[n]$: impulse response, also called a filter kernel.
- $y[n]$: output signal.
- The impulse response is the system's complete fingerprint for LTI behavior.



A delta input reveals $h[n]$; every other input is assembled from shifted and scaled deltas.

Delta Functions: Identity, Gain, and Delay

Three basic kernels

$$x[n] * \delta[n] = x[n]$$

$$x[n] * k\delta[n] = kx[n]$$

$$x[n] * \delta[n - s] = x[n - s]$$

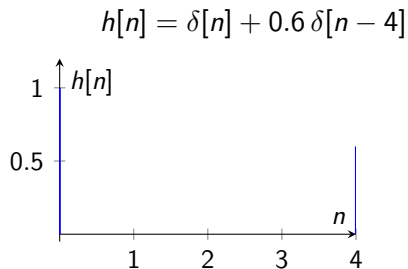
- $\delta[n]$ is the identity element of convolution.
- Scaling the delta creates amplification or attenuation.
- Shifting the delta creates delay or advance.
- These model ideal storage, communication, and measurement.

Echoes as Multi-Delta Impulse Responses

A direct sound plus one delayed reflection

$$h[n] = \delta[n] + a \delta[n-D] \implies y[n] = x[n] + a x[n-D].$$

- D is the delay in samples.
- a is the echo amplitude relative to the original.
- Audio reverberation, radar, sonar, seismic exploration, and telephone echo suppression all depend on this idea.
- Multiple echoes are modeled by adding more shifted delta taps.



Calculus-Like Operations

First difference: discrete derivative

$$h_{\Delta}[n] = \delta[n] - \delta[n - 1]$$
$$y[n] = x[n] - x[n - 1].$$

- Highlights changes, slopes, edges, and transitions.
- A symmetric option is
$$\frac{x[n + 1] - x[n - 1]}{2}.$$

Running sum: discrete integral

$$h_{\Sigma}[n] = u[n]$$
$$y[n] = x[n] + y[n - 1] = \sum_{k=-\infty}^n x[k].$$

- Accumulates area under the sequence.
- It is recursive and has an infinite impulse response.

Important distinction

These are discrete analogues of derivative and integral operations; the formal calculus terms belong to continuous-time signals.

Implementation View: Difference Equations

First difference

$$y[0] = 0, \quad y[n] = x[n] - x[n-1].$$

$$y[0] = 0$$

for $n = 1..N-1$:

$$y[n] = x[n] - x[n-1]$$

Running sum

$$y[0] = x[0], \quad y[n] = y[n-1] + x[n].$$

$$y[0] = x[0]$$

for $n = 1..N-1$:

$$y[n] = y[n-1] + x[n]$$

- A full convolution routine is unnecessary when the kernel has this simple recursive or local structure.
- Difference equations are the language of many digital filters.

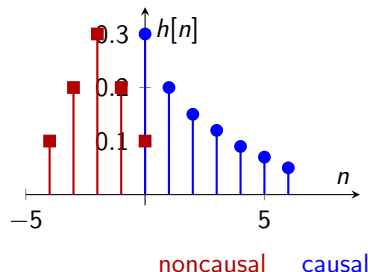
Causal and Noncausal Signals

Causality condition

A discrete-time system is causal if an input at time n cannot affect outputs before time n .

$$h[n] = 0 \quad \text{for all } n < 0.$$

- Real-time physical systems must be causal.
- Offline DSP can be noncausal because the entire signal array is already available.
- A finite noncausal kernel can often be made causal by adding delay.



Zero Phase, Linear Phase, and Nonlinear Phase

Zero phase

Symmetric about $n = 0$.

$$h[n] = h[-n]$$

The phase spectrum is zero.

Linear phase

Symmetric about a nonzero center $n = n_0$.

$$h[n_0 + m] = h[n_0 - m]$$

The phase spectrum is a straight line.

Nonlinear phase

No left-right symmetry. Different frequency components experience different relative delays.

Why this matters

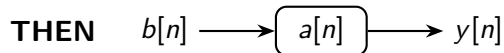
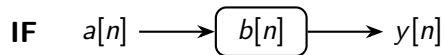
Linear-phase FIR filters preserve waveform shape better because frequency components are delayed by a common group delay.

Commutative Property

Order does not change the result

$$a[n] * b[n] = b[n] * a[n].$$

- Mathematically, either signal can be treated as the input or as the impulse response.
- Physically, this exchange may not make sense even though the algebra is valid.
- Useful for rearranging DSP expressions and simplifying proofs.



Associative Property: Cascaded Systems

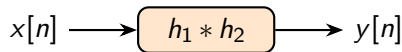
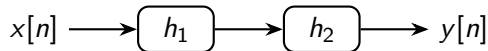
Grouping does not change the result

$$(a[n] * b[n]) * c[n] = a[n] * (b[n] * c[n]).$$

- Cascaded systems may be rearranged without changing the final output.
- Several cascaded systems can be replaced by one equivalent:

$$h_{\text{eq}}[n] = h_1[n] * h_2[n] * \dots * h_K[n].$$

- Foundational for multistage filtering, equalization, and block-diagram simplification.

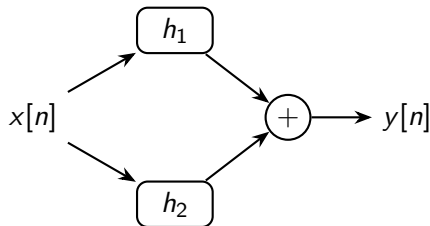


Distributive Property: Parallel Systems

Convolution distributes over addition

$$a[n] * b[n] + a[n] * c[n] = a[n] * (b[n] + c[n]).$$

- Parallel systems sharing the same input and summed at the output can be merged.
- Equivalent impulse response:
 $h_{\text{eq}}[n] = h_1[n] + h_2[n]$.
- Useful in filter banks, echo cancellation, and systems built from additive branches.



$$h_1 + h_2$$

Transference Between Input and Output

A linear change before the system appears after the system

If $y[n] = x[n] * h[n]$, and the input is changed by another LTI operation $g[n]$,

$$x'[n] = x[n] * g[n],$$

then

$$y'[n] = x'[n] * h[n] = x[n] * g[n] * h[n] = y[n] * g[n].$$

- Amplify the input by 2: the output is amplified by 2.
- Difference the input: the output is differenced.
- Filter the input: the output is filtered in the same way.

Correlation: Detecting a Known Waveform

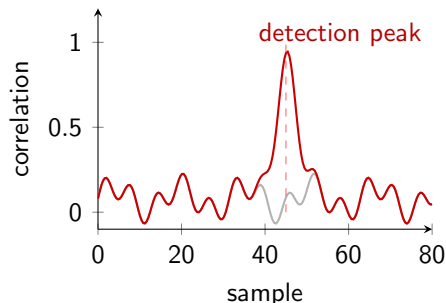
Problem

Given a known target waveform $t[n]$, determine where it occurs inside a received signal $x[n]$ that also contains noise.

Cross-correlation

$$r_{xt}[\ell] = \sum_n x[n] t[n - \ell].$$

- A large positive peak means the target aligns well at lag ℓ .
- Core operation behind radar, sonar, matched filtering, and template matching.
- Random noise creates correlation noise,



Convolution vs. Correlation

Convolution

$$y[n] = x[n] * h[n] = \sum_k x[k] h[n - k].$$

- Describes how an LTI system transforms a signal.
- One signal is effectively flipped during the operation.

Correlation

$$r_{xt}[\ell] = \sum_n x[n] t[n - \ell].$$

- Measures similarity as a function of shift.
- No left-right flip of the target in the basic machine view.

Connection

Correlation can be computed with convolution by preflipping one signal:

$$\text{corr}(a, b) \equiv a[n] * b[-n].$$